

Michelle Adu-Asah

ITAI-2373 — NLP Final Project

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Reflective Journal — NewsBot 2.0

Just like my midterm, I chose to work on this final project by myself. Working individually allows me to move at my own pace, stay organized the way I want to, it also makes me learn all the areas better and pushes me to research and gain deeper knowledge.

The hardest part of this project was writing and debugging the code. I ran into several errors throughout the process, especially when working with libraries I had not used before. There were moments where something that looked straightforward on paper just would not run the way it was supposed to. I ended up using Gemini and Claude Ai to help me work through some of those errors, which actually turned out to be a useful experience in itself.

Out of everything I added in this final project, the multilingual feature was honestly my favourite part to build. Getting the system to detect a language and then translate text into German, French, and Spanish felt very personal to me because I am multilingual myself. I know firsthand how different languages carry meaning in different ways and how translation is not always a straightforward swap of words. That background made this module feel more meaningful than just a technical requirement.

This project taught me a lot more than I expected, and not just technically. The biggest thing I took away was learning how to figure things out as I go. When you run into an error using a library you have never touched before, there is no shortcut — you have to read, test, and keep trying until you get it. I got a lot more comfortable working with NLP tools like sentiment analysis, topic modeling, and named entity recognition throughout this process. Each module adds its own layer of understanding to the text, and seeing them all work together in one system gave me a much clearer picture of how NLP pipelines are structured in practice.

If I had more time I would improve several areas of the system. The first would be the user interface. Right now the backend works and the outputs are correct, but the way a user actually interacts with it is still basic. I would want to turn it into a proper web app where someone can paste an article and see the results laid out clearly and visually.

I would also upgrade the models. Most of what I used are traditional machine learning or rule-based approaches, which work well for this level of project but are not as powerful as modern deep learning models. Replacing the classifier and sentiment analyzer with something like BERT would likely improve accuracy quite a bit.

The chatbot is another area I would revisit. Right now it works by matching keywords, which means if someone phrases a question slightly differently it might not respond correctly. I would want to make it smarter so that it can understand the intent behind a question rather than just looking for specific words.

The multilingual feature, while my favourite, still has limitations. The translations are not always perfectly accurate and the current translator has a character limit, which means longer texts get cut off. I would want to use a more accurate translation model and expand the language support beyond the four I currently have.

Finally, I would work on the dataset. The system was trained on BBC news articles only, which means it may not perform as well on different types of content. Training on a larger and more diverse dataset would help it generalize better to real-world scenarios.

Building this project genuinely changed how I look at tools I use every day. Take Google Translate — I have always known it is not perfect, but now I understand exactly why. It is honestly fine for single words, but when it comes to full sentences it often misses the context and tone that a fluent speaker would catch immediately. Having built a translation feature myself, I now see the technical limitations behind it rather than just noticing when the output sounds off.

Chatbots are another one. I used to interact with them as just a user — type something in, get something back. Now I think about what is happening underneath. Is it keyword matching like mine? Does it have actual language understanding? How is it deciding what to say? Building the conversational interface for this project, as simple as it is, made me much more aware of the gap between a basic rule-based system and something that actually understands language. It gave me a real appreciation for how much goes into making something that feels natural to use.

This project was the most complete thing I have built in this course. Going from the midterm, where I had separate modules that mostly worked on their own, to a final system where everything connects and runs together through one function — that felt like real progress. It was not always easy and there were definitely moments of frustration, but looking back I am glad I pushed through each one because that is where most of the actual learning happened. I came into this course understanding NLP from a theoretical perspective, and I am leaving it with a much better sense of what it takes to actually build something with it, especially our classes which helped me significantly by the simplicity and relatability.